



XXV MEETING OF PHYSICS

Pneumonia Classification from Chest X-Rays Using a Quantum Spiking Neural Network

Francis Lluen¹, Paul Reyna¹, Piero Romero³, Pedro Rojas^{2,6}, Enver Oruro⁵, Ricardo Quispe^{2,6}

¹Universidad Nacional de Ingeniería ²Universidad Nacional Mayor de San Marcos ³Pontificia Universidad Católica del Perú

⁵UNESCO UniTwin Complex Systems Digital Campus ⁶Quantumquipu

Lima, Perú 2026



Meeting of
PHYSICS

Abstract

The increasing use of neural networks in medical image analysis has motivated the exploration of alternative computational paradigms, including Quantum Spiking Neural Networks (QSNNs). These models combine quantum computing principles with the temporal processing characteristics of spiking neurons [1]. In this work, we explore their application to chest X-ray classification using the PneumoniaMNIST dataset [2].

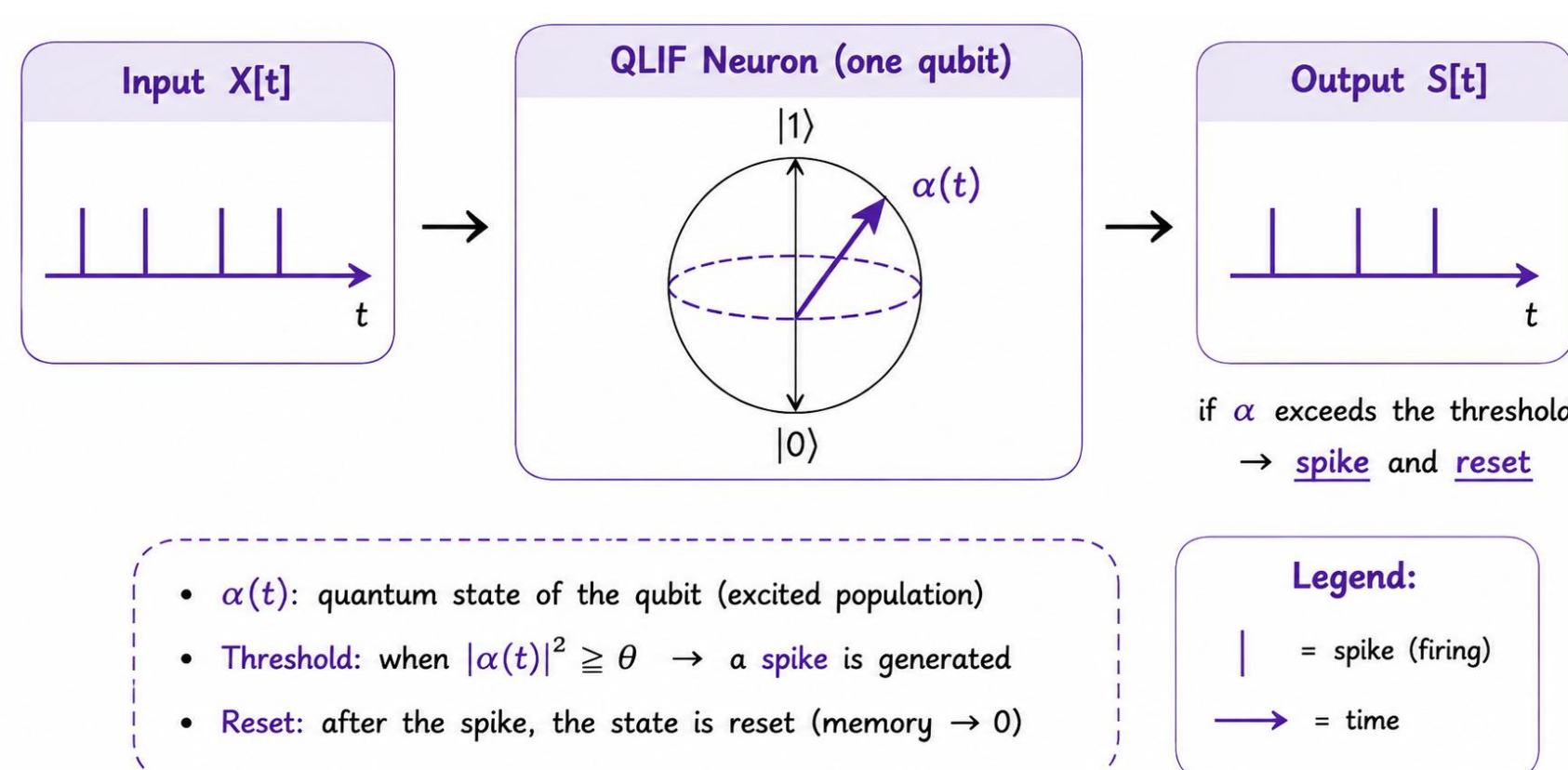
A previously published QSNN architecture based on the QLIF neuron is implemented with a proposed memory formulation that introduces the expected exponential decay of the neuronal state. The network is trained to classify chest X-ray images into “Pneumonia” and “Normal”, and its performance is evaluated using a test set. The evolution of the training loss, classification results, and neuronal memory dynamics are analyzed to characterize the behavior of the proposed model.

The results indicate that the proposed QSNN is capable of learning meaningful distinctions between the two classes. This study establishes a reproducible basis for further investigation of QSNNs in image classification, including evaluations across multiple random seeds and future experiments involving quantum noise models and real hardware implementations.

INTRODUCTION

Quantum Spike Neural Networks (QSNNs) combine quantum computing with biological time processing. In this work, we reproduce and evaluate a QSNN based on quantum neurons with Leaky Integrate-and-Fire (QLIF) dynamics and a proposed exponential decay, to classify chest X-ray images from the PneumoniaMNIST dataset [2] into the 'Normal' and 'Pneumonia' classes. This allows us to analyze its learning dynamics and characterize the quantum neuron.

BEHAVIOR OF A QLIF NEURON



From [1] we obtained an equation to a classical neuron integrate and fire (LIF)

$$U[t] = \beta U[t-1] + (1-\beta)I[t]$$

Where, β is a decay factor and I the input current. If we want to transform a LIF in a QLIF, we express the potential U like

$$\alpha[t] = P(|1\rangle) = \sin^2(\theta/2)$$

INPUT SPIKE

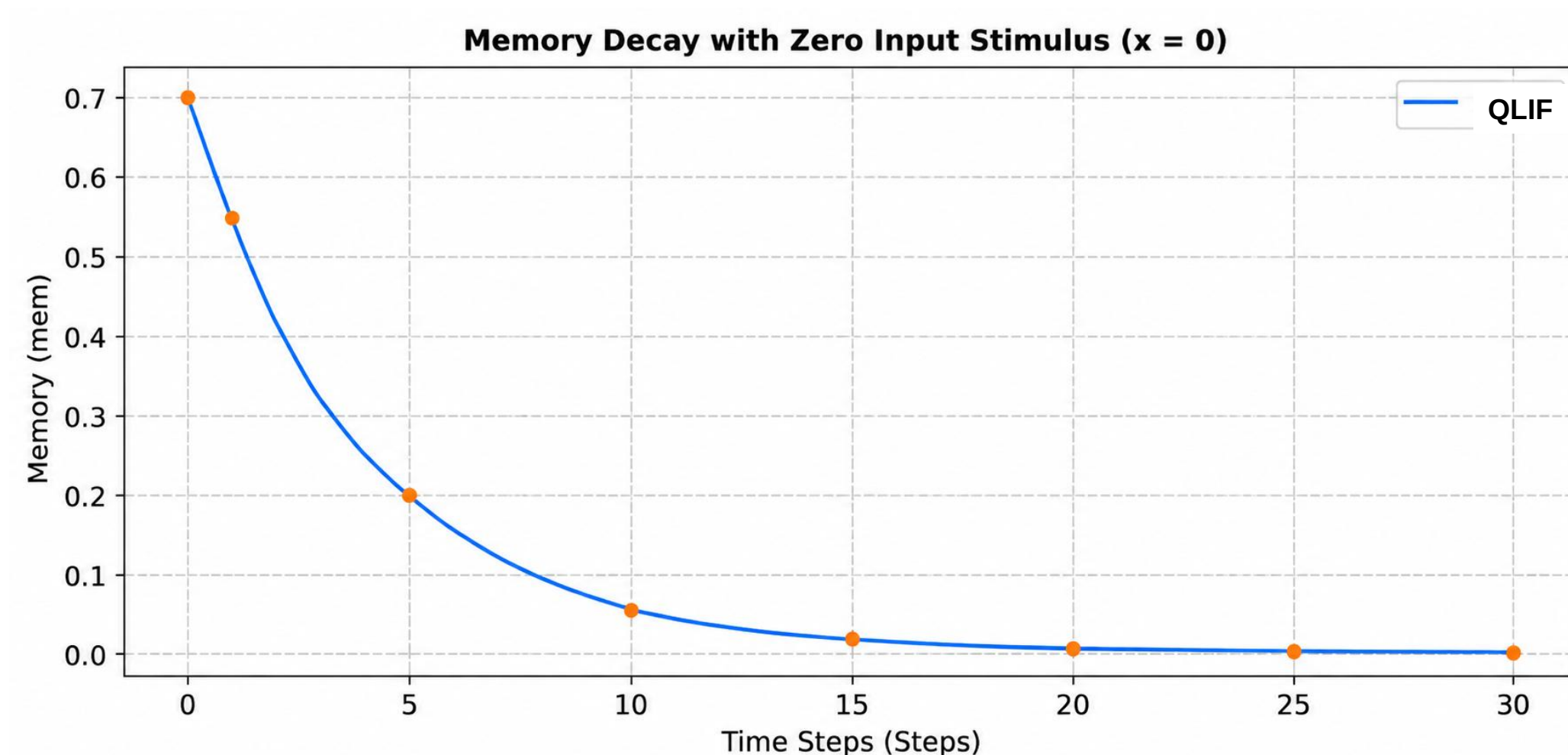
When our neuron receives an impulse, the value of α increases, thus facilitating reaching the neuron's firing threshold. This increase in the excited population when there is an input spike ($x=1$) is represented in [1] by

$$\alpha[t+1] = \sin^2\left(\frac{\theta + \varphi(t)}{2}\right), \text{ with } \varphi(t) = 2 \arcsin \sqrt{\alpha[t]}$$

PROPOSED DECAY

Our QLIF neuron shows an exponential decrease in its internal memory in the absence of external stimuli ($x=0$). This formulation, which has been an alternative proposal based on paper [1], allows the network to retain temporal information from previous stimuli and simulate a biological leakage behavior, the decrease is represented by

$$\alpha[t+1] = \sin^2\left(\frac{\gamma(t) + \varphi(t)}{2}\right), \text{ with } \gamma(t) = 2 \arcsin \sqrt{\alpha(t)e^{\tau/T_1}} - \varphi(t)$$

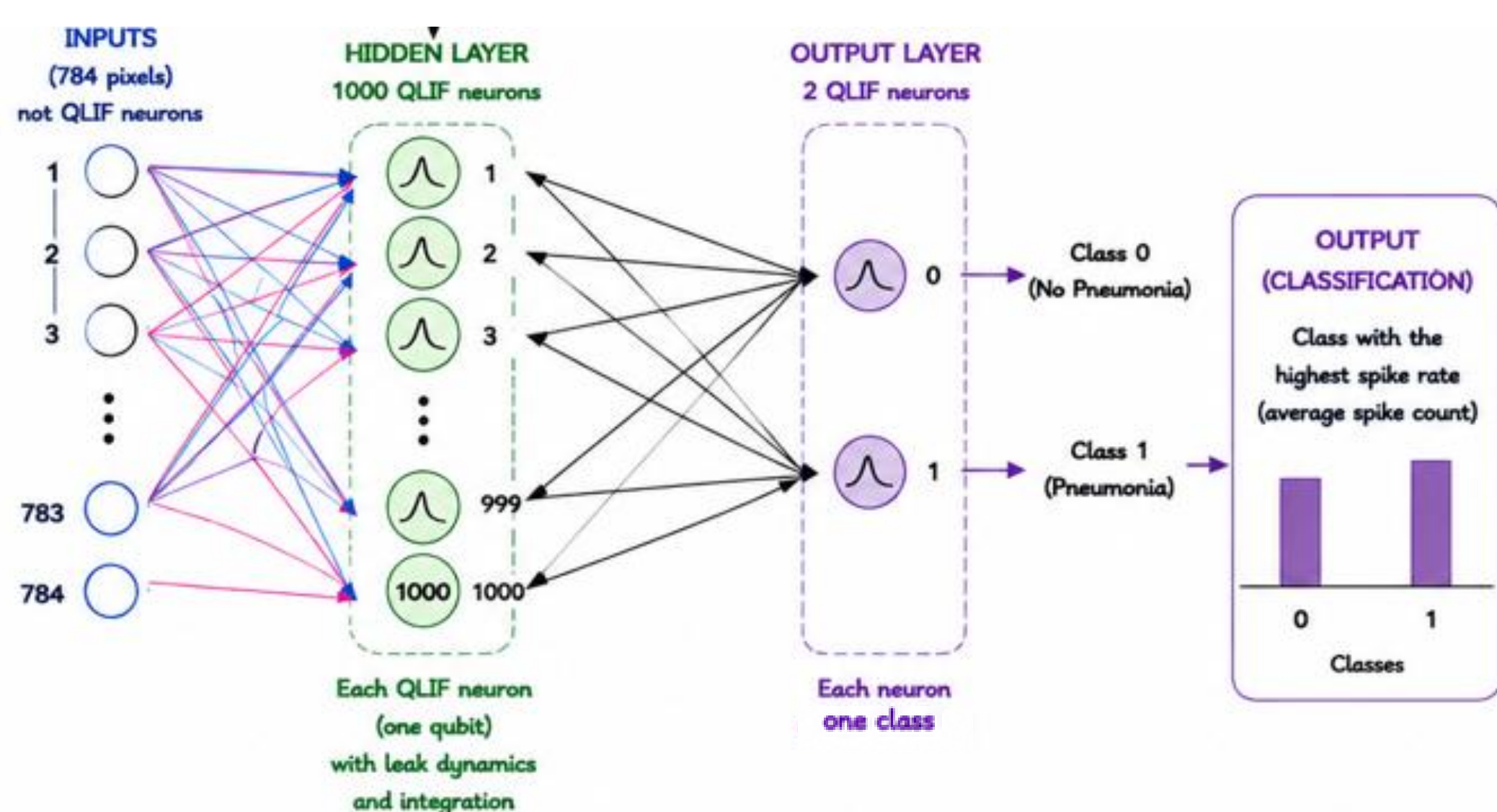


STRUCTURE OF THE NEURAL NETWORK

We implemented the QSNN in Python using PyTorch on the PneumoniaMNIST dataset, which consists of 3,882 training images and 624 test images (28x28 pixels grayscale, normalized on [0,1]). The architecture comprises an input layer of 784 neurons, a hidden layer of 1,000 QLIF neurons (Alpha), and an output layer of 2 QLIF neurons. Training was performed over 5 epochs (batch size 200, Adam optimizer with a rate of 5e-3, and weighted CrossEntropy loss to compensate for imbalance), updating the θ and τ parameters that feed back into the network at each iteration. The Loss is calculated by the backpropagation through time method (BPTT) described by

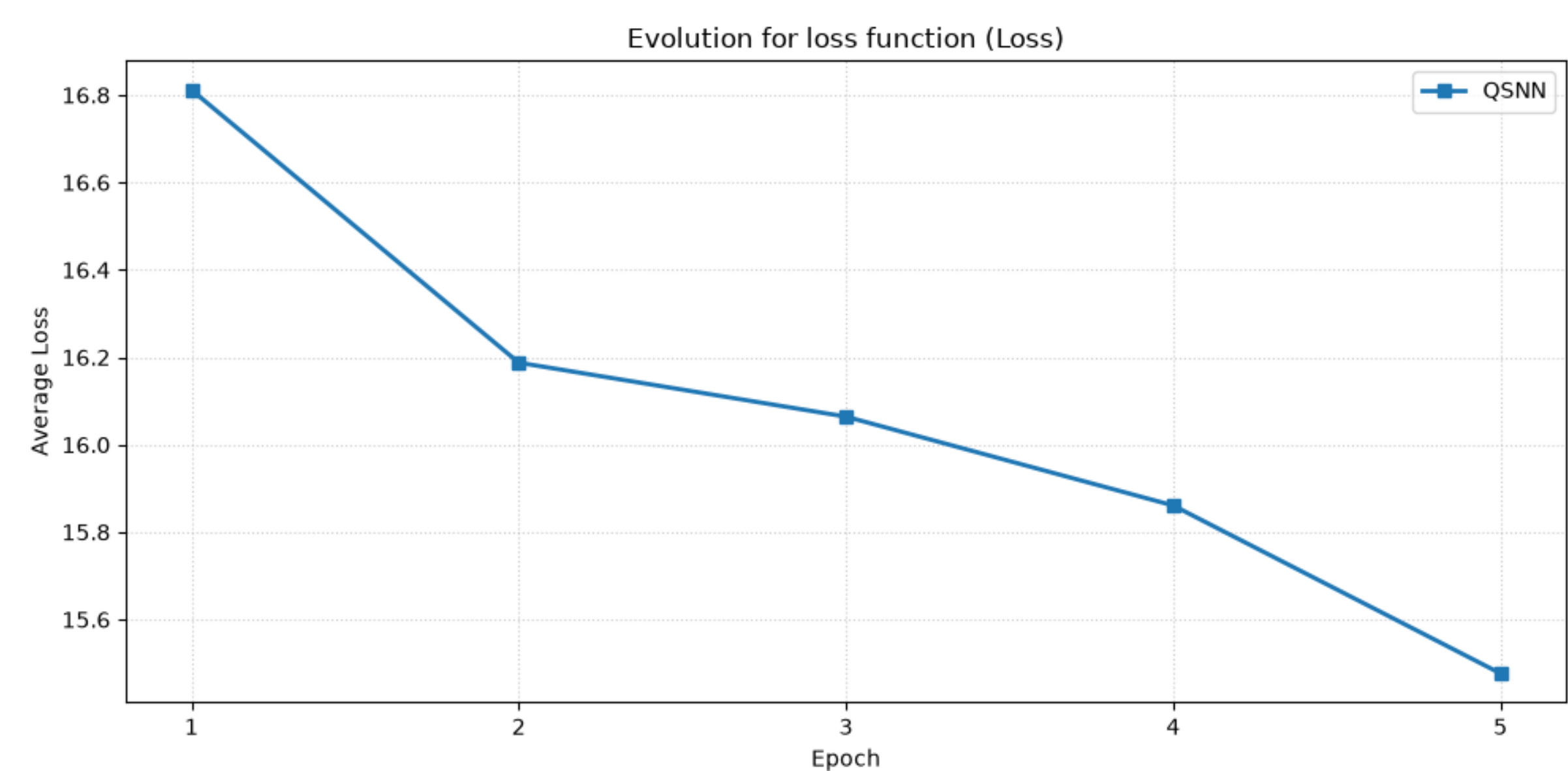
$$\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial S} \frac{\partial S}{\partial \alpha} \frac{\partial \alpha}{\partial W}, \text{ where } \frac{\partial S}{\partial \alpha} = \frac{1}{\pi} \frac{1}{1 + (\alpha[t]\pi)^2}, \text{ and } W = \begin{cases} W = W_\theta \\ \text{or} \\ W = W_\tau \end{cases}$$

Finally, classification was determined by assigning the sample to the output neuron with the highest number of spikes generated over 25 time steps.

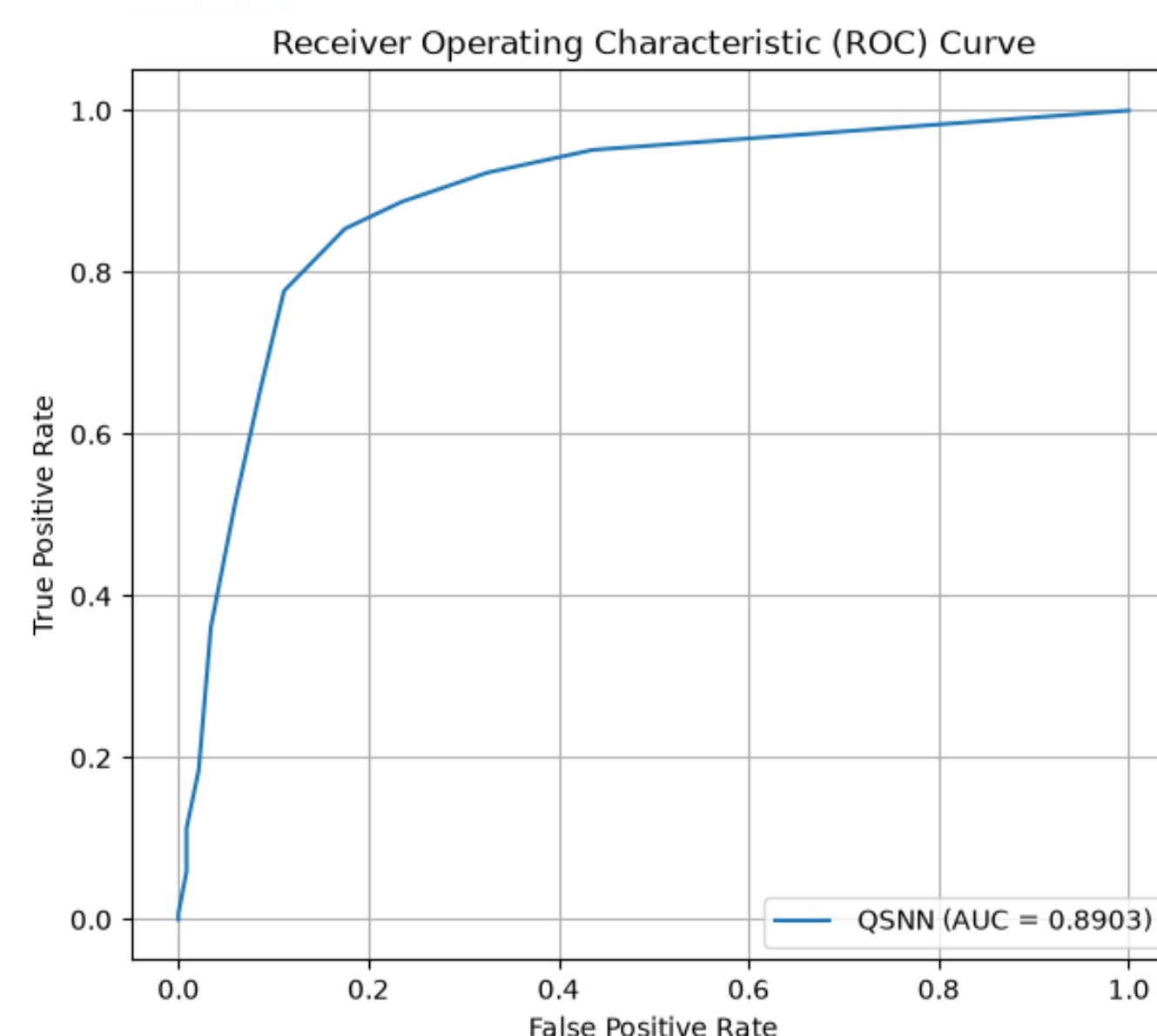
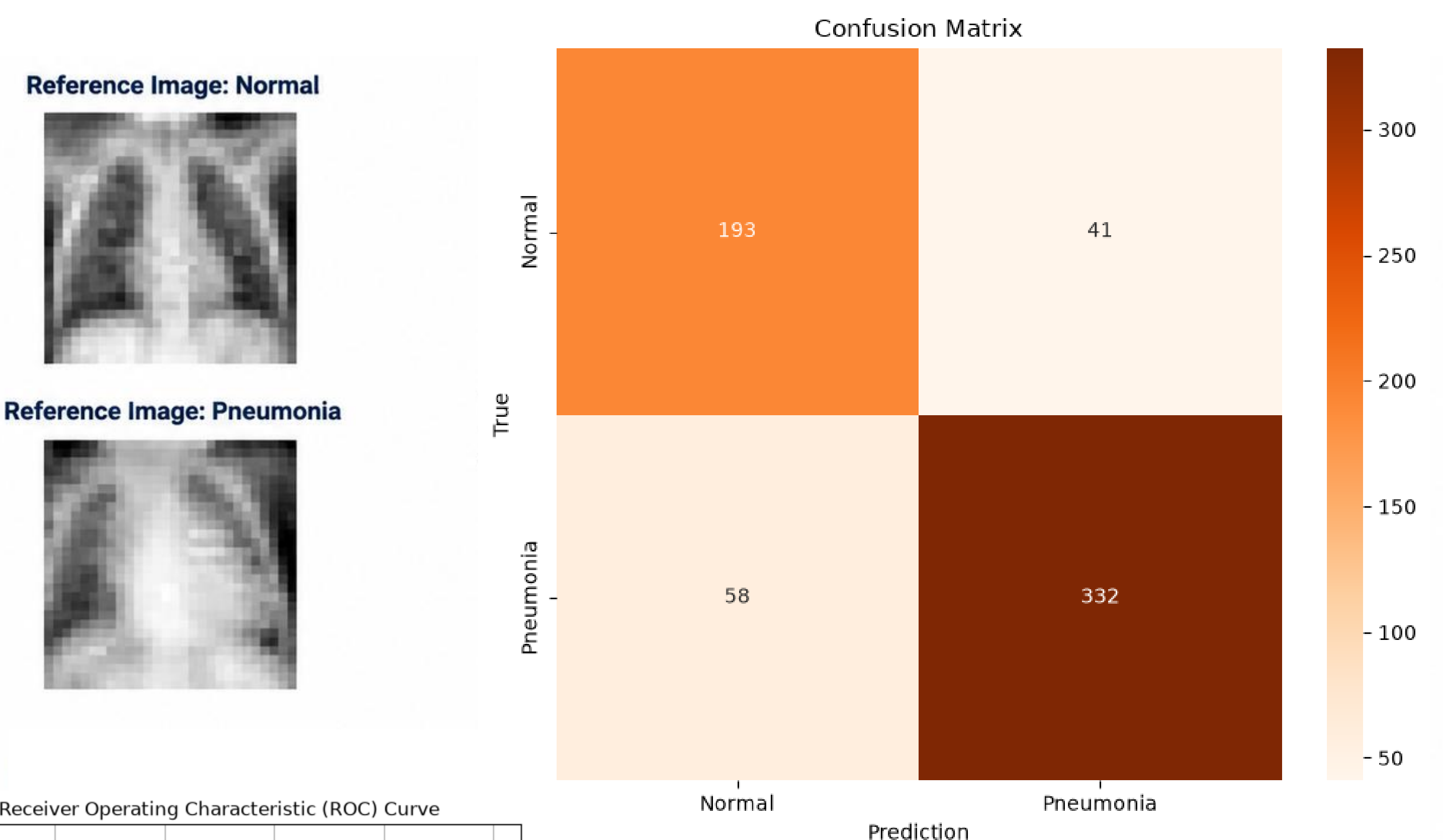


RESULTS

The average loss during the five epochs showed a continuous decreasing trend, with values of 16.81, 16.19, 16.06, 15.86, and 15.48. This reduction of 1.33 units (approximately 7.9% from the initial value) confirms that the optimization algorithm correctly adjusted the network parameters to minimize the cost function over the training set.



The model achieved an accuracy of 84.13%. Complementary weighted metrics were: precision of 0.8446, recall of 0.8413, and F1-score of 0.8424. These values, close to each other, indicate that the classifier maintains balanced performance in both categories, without pronounced bias toward either class. The confusion matrix (rows = actual label, columns = predicted label) breaks down the hits and misses: the Normal class had 193 hits versus 41 false positives, while the Pneumonia class had 332 hits and 58 false negatives. This behavior reflects greater sensitivity for the majority class because there are more Pneumonia images for the model to learn.



	Metrics	Value
0	Accuracy	84.13%
1	Precision (weighted)	0.8446
2	Recall (weighted)	0.8413
3	F1-Score (weighted)	0.8424

CONCLUSION

In this work, a QLIF neuron with exponential memory decay in the absence of a stimulus was proposed, and its implementation achieved an accuracy of 84.13% on the test set. The resulting confusion matrix reflects 193 correct classifications for the Normal class and 332 for the Pneumonia class, while the cross-classification errors amounted to 41 and 58, respectively. Furthermore, the loss function showed a decreasing trend over the five epochs, decreasing from 16.81 to 15.48, which demonstrates the convergence of the model and the effectiveness of the parametric adjustment during training, consistent with the findings reported in [1]. It should be noted that these results correspond to a specific execution of the Python code and should therefore be interpreted as a case study and not as a generalized estimate of the classifier's average performance.

References

- [1] D. Brand and F. Petruccione, "A quantum leaky integrate-and-fire spiking neuron and network," *npj Quantum Information*, vol. 10, Art. no. 129, 2024, doi: 10.1038/s41534-024-00921-x.
- [2] Rijulshr, "PneumoniaMNIST," Kaggle, 2024. [Online]. Available: <https://www.kaggle.com/datasets/rijulshr/pneumoniamnist>.